Week 6: Poisson regression on Ashford matches

Decision record prepared and analysis run: 12 November 2026. The original PC-session date was not supplied. Data: fictional Midland Premier League, 2023/24.

Aim and choice of method

We used Poisson regression to investigate whether Ashford United's goals scored are associated with home/away status, the opponent's final rank and goals conceded. This develops the Week 2 normal linear regression using a distribution designed for non-negative integer counts.

The project data brief assigns **Poisson regression to the Week 6 practical**. The combined Weeks 6 and 8 lecture notes introduce contingency tables for subsequent practical sessions. We therefore followed the Week 6 practical's count-model analysis, link comparison, deviance tests, exposure discussion and diagnostics; categorical contingency-table analyses are left for Weeks 8 and 9.

1. Data preparation and initial model

All **46 matches** were retained: **23 home and 23 away**. The four analysis variables have no missing values. Ashford scored **60 goals**, with a mean of **1.304** per match and **14 zero-goal matches**. The home mean was **1.739** and the away mean **0.870**. Row 8 is an **8-0 home win against the opponent ranked 24th** and was flagged for investigation, not deleted.

We made `home_away` a factor with **Away as the reference**. We treated `opponent_rank` as numeric: a larger number indicates a lower league position. We created `rank_c = opponent_rank - 10`, making the intercept interpretable at rank 10. This centring changes the intercept but not the fitted means or rank slope. A constant effect per rank place on the log scale is a modelling assumption.

The initial model was:

```
glm(goals_scored ~ home_away + rank_c + goals_conceded,
    family = poisson(link = "log"), data = matches)
```

Writing H = 1 for Home and 0 for Away, R = opponent rank and C = goals conceded:

$$Y_i \mid H_i, R_i, C_i \sim \text{Poisson}(\mu_i)$$
$$\log(\mu_i) = \beta_0 + \beta_H H_i + \beta_R (R_i - 10) + \beta_C C_i.$$

The log link gives positive fitted means. A Poisson model assumes that conditional variance equals conditional mean; the raw variance of **1.994** exceeding the overall mean does not by itself establish overdispersion after adjustment.

Exposure decision: the unit is one league fixture. There is no varying exposure measure in the match data, so no offset is needed. An offset of `log(1)` for each match would be zero. Opponent rank is a predictor, not an exposure.

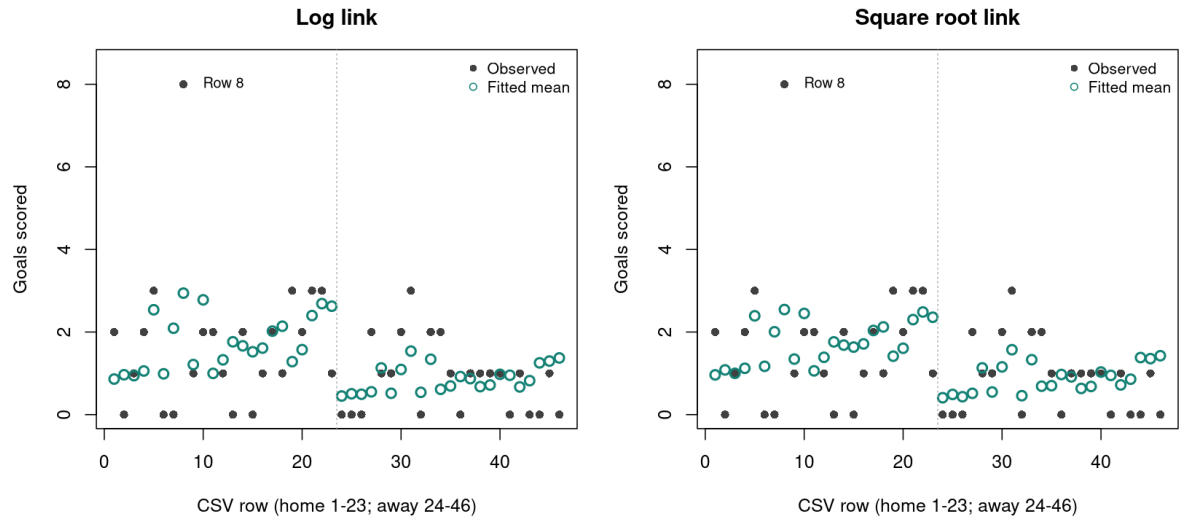
Interpretation decision: goals conceded is another outcome of the same match, and final rank is measured at season end. These models describe associations in this season; they are not validated pre-match forecasts or causal estimates.

2. Comparing the log and square-root links

We fitted the same additive predictors to the same 46 observations under both links.

Link	Residual deviance	Residual df	AIC
Log	49.644	42	135.292
Square root	50.734	42	136.382

Decision: retain the log link. It has slightly lower deviance and AIC and gives an uncomplicated multiplicative interpretation. The AIC difference is only **1.09**, so it is a modest preference, not decisive evidence against the square-root link. These models are not nested, so a chi-square test of their deviance difference is inappropriate.



Both models give similar fitted means and substantially underpredict the eight-goal match. The horizontal axis is **CSV row order**, with home fixtures followed by away fixtures; it is not a time axis. For the alternative link, `sqrt(mu) = eta` requires non-negative eta. Its coefficients must not be exponentiated and interpreted as constant goal ratios.

3. Model selection using differences in deviance

We checked a home-by-rank interaction, then considered removing goals conceded. We used nested-model likelihood-ratio tests at the **5% level**, retaining main effects when considering an interaction, as required by model hierarchy.

Comparison	Increase in deviance on removal	df	LRT p-value	Decision
Remove home-by-rank interaction, keeping goals conceded	0.302	1	0.583	Use additive effects
Remove goals conceded from additive model	0.140	1	0.708	Remove goals conceded
Remove home/away from selected model	6.796	1	0.00914	Retain home/away
Remove rank from selected model	8.263	1	0.00405	Retain rank

Removing goals conceded reduces AIC from **135.292 to 133.433**, leaving home/away status and centred opponent rank as the selected predictors.

The improvement over the intercept-only model is **15.059 on 2 df, p = 0.000537**. Non-significance does not prove an association is zero. These findings require the influence check below.

4. Selected model and interpretation

$$\log(\widehat{\mu}_i) = -0.355324 + 0.693147 H_i + 0.054348(R_i - 10).$$

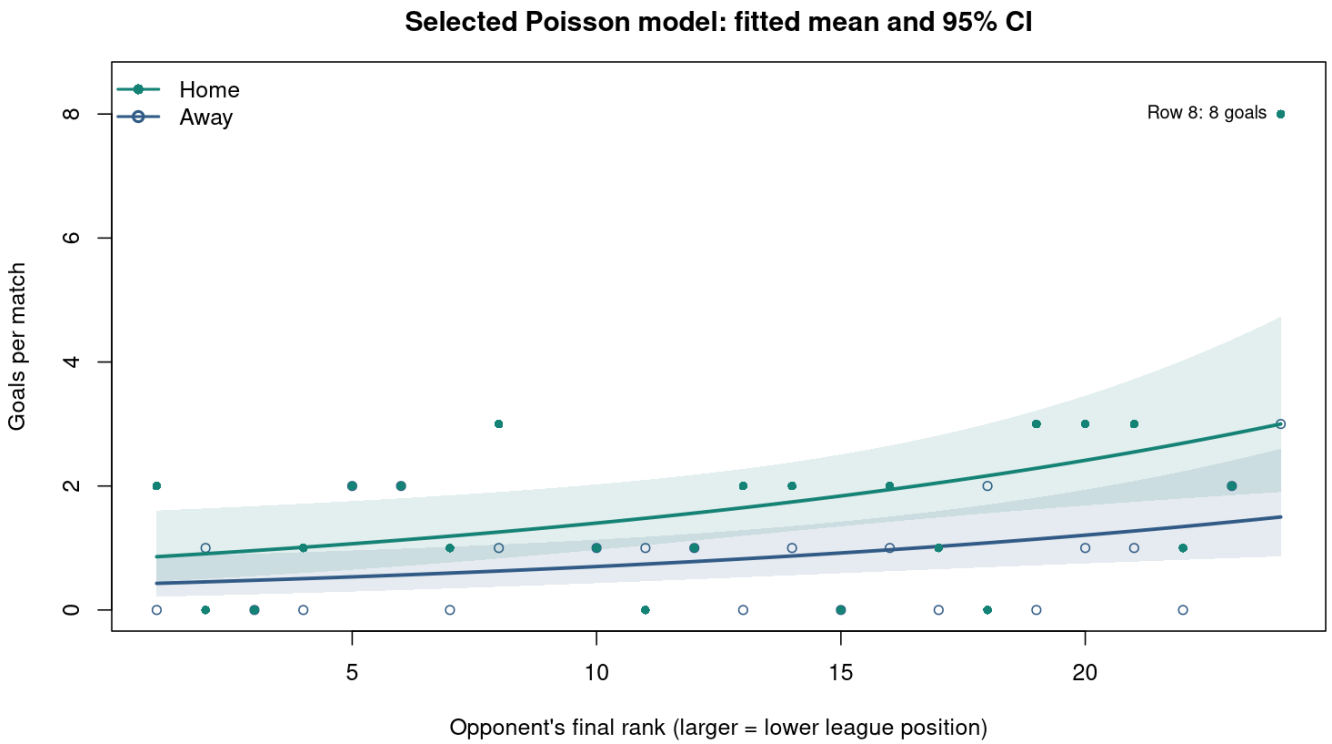
Term	Estimate	Standard error	exp(coef.)	95% Wald CI for exp(coef.)	Wald p-value
Intercept	-0.3553	0.2456	0.701	0.433-1.134	0.148
Home versus Away	0.6931	0.2739	2.000	1.169-3.421	0.0114
Rank, one place larger	0.05435	0.01939	1.0559	1.0165-1.0968	0.00507

Home/away: at the same opponent rank, the fitted mean goals scored at home is **2.00 times** the away mean, a 100% increase. This is a ratio of expected counts per match, not an odds ratio or a fixed increase of one goal.

Opponent rank: holding venue fixed, a one-place increase in numerical rank is associated with approximately **5.6% more expected goals**. This interpretation assumes a linear rank effect on the log scale.

Intercept: `exp(-0.355324) = 0.701` is the expected number of goals in an away match against a rank-10 opponent. It is an expected count, not a ratio.

Opponent rank	Expected goals Away	Expected goals Home
1	0.430	0.860
10	0.701	1.402
24	1.500	3.000

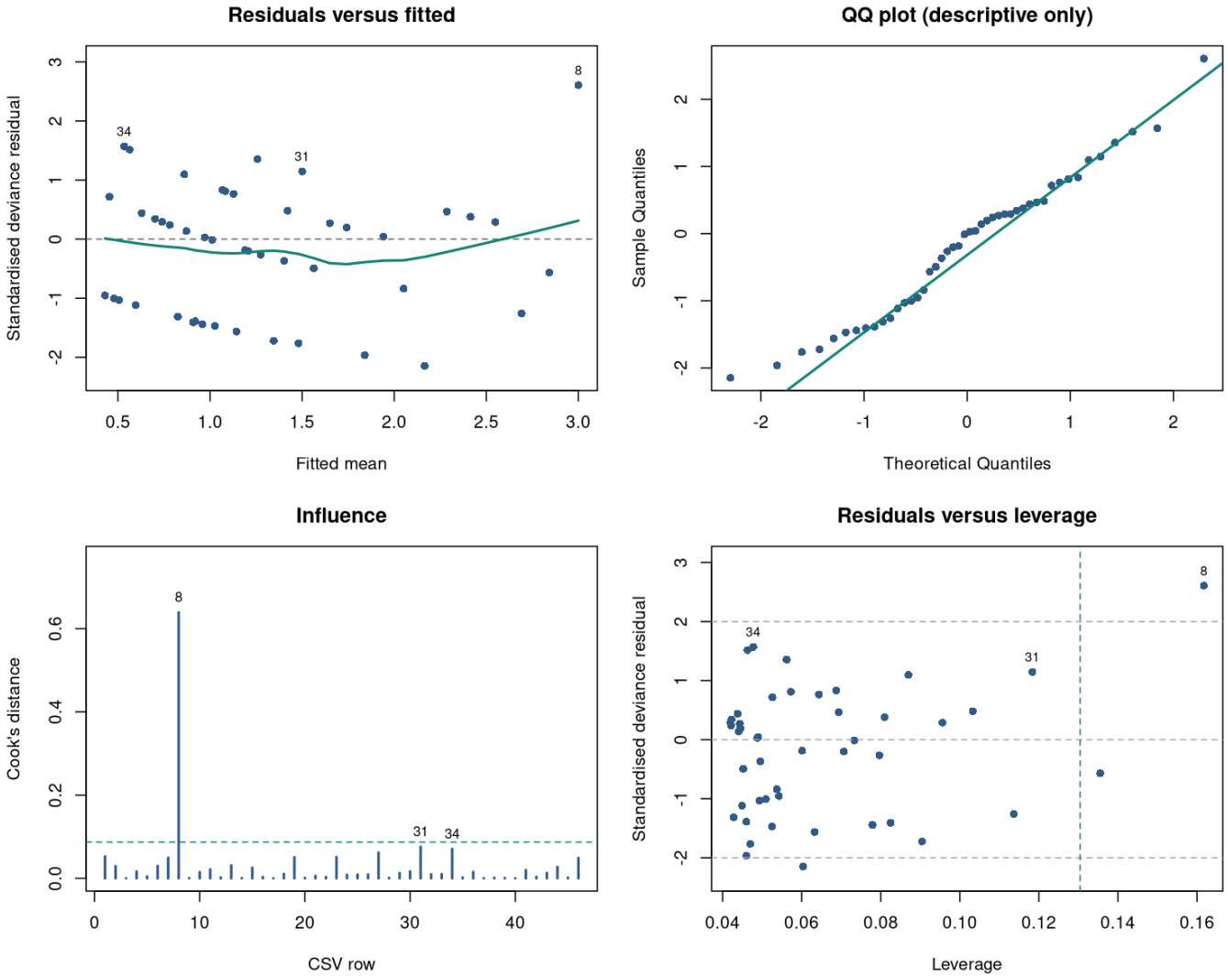


Lines show fitted means; shaded regions are **95% confidence intervals for those means**, calculated on the log scale and transformed back. They are not prediction intervals for individual match scores. The home mean is twice the away mean throughout because the selected model has no interaction.

5. Goodness of fit and diagnostic plots

The selected model's residual deviance is **49.785 on 43 df**. The practical's approximate chi-square goodness-of-fit test gives **p = 0.221**, so it does not detect lack of fit at 5%. Fitted means range from only **0.430 to 3.000**, so the chi-square approximation is imperfect; this result is not proof of correct fit.

The Pearson statistic is **42.588**, giving estimated dispersion **42.588 / 43 = 0.990**. There is no clear residual overdispersion signal here, so we retained the ordinary Poisson model. The line “dispersion parameter ... taken to be 1” in R's summary describes the model assumption, not a diagnostic test. The model also expects about **15.00 zero-goal matches**, compared with **14 observed**; this is a descriptive check, not a formal zero-inflation test.



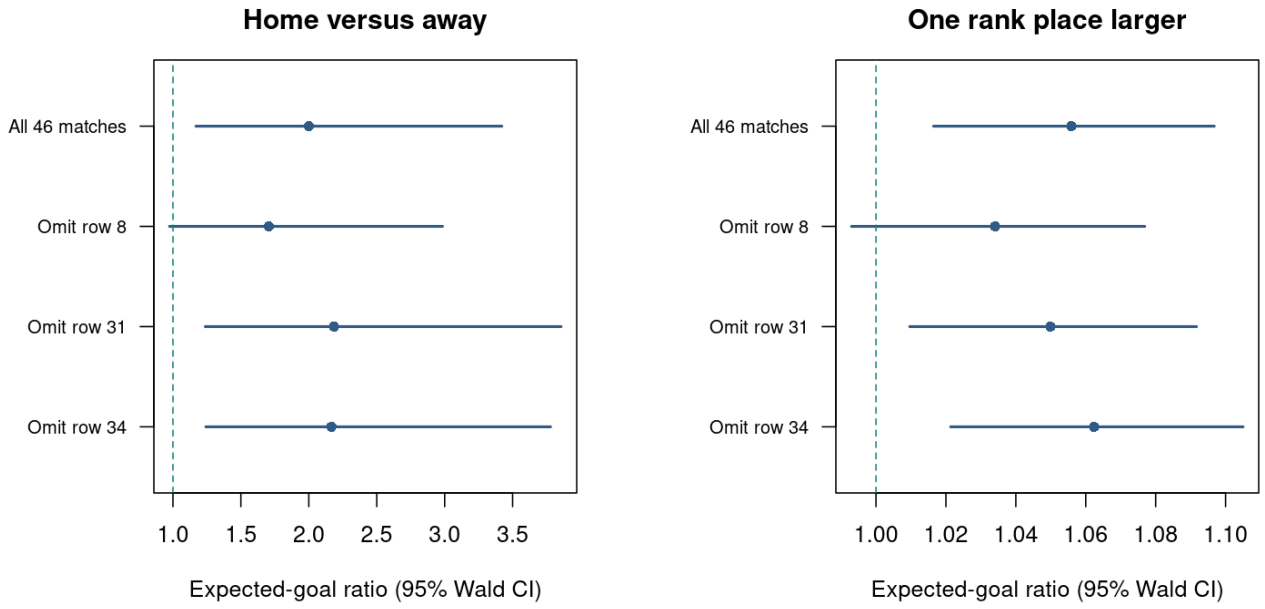
The residual plot has a mild curved smooth and the discrete bands expected with low counts. It does not establish a perfectly specified mean function. The QQ plot has some tail departures, but normal errors are not a Poisson-model assumption. The clearest diagnostic issue is **row 8**, with fitted mean **3.000**, standardised deviance residual **2.606**, Cook's distance **0.639** and leverage **0.162**.

Only row 8 exceeds the Cook's-distance reference **4/46 = 0.087**. Rows **8 and 10** exceed the leverage reference **2 x 3 / 46 = 0.130**. These are investigation thresholds, not automatic deletion rules; row 10 has low Cook's distance despite its high leverage.

6. Sensitivity to influential matches

We refitted the selected formula after omitting each of the three largest Cook's-distance observations separately. All other matches were retained in each refit. The full-data fit remains the main analysis.

Data used	Home/Away goal ratio (95% Wald CI)	Home LRT p	Rank goal ratio (95% Wald CI)	Rank LRT p
All 46 matches	2.000 (1.169-3.421)	0.00914	1.0559 (1.0165-1.0968)	0.00405
Omit row 8	1.706 (0.975-2.984)	0.0576	1.0341 (0.9930-1.0769)	0.102
Omit row 31	2.185 (1.238-3.857)	0.00493	1.0499 (1.0096-1.0917)	0.0127
Omit row 34	2.167 (1.242-3.779)	0.00466	1.0624 (1.0214-1.1050)	0.00186



Decision: retain row 8 because no recording error is established and an eight-goal match is possible. However, its omission reduces both estimated effects and moves both LRT p-values above 0.05. The direction of the associations remains positive, but their statistical significance is sensitive to this single result. Omitting rows 31 or 34 does not have that consequence. We should not describe the original significance findings as robust.

7. Decisions recorded and limitations

Decision	Reason
Model goals scored with a Poisson GLM	Goals are counts; a normal model is not the natural starting distribution.
Use log link, Away reference and rank centred at 10	Positive fitted means and interpretable coefficients and baseline.
Use one fixture as exposure, without a varying offset	No varying match exposure is supplied.
Keep home/away and rank; exclude goals conceded and the home-by-rank interaction	Supported by the stated nested comparisons, AIC and parsimony.
Retain all 46 matches and report influence sensitivity	There is no verified reason to delete the influential 8-0 result.
Keep the ordinary Poisson variance assumption for now	Pearson dispersion is close to 1; diagnostics do not show clear overdispersion.
Preserve every candidate model's code and outputs	The analysis decisions can be checked and revisited in later weeks.

Only one simulated season is observed. Conditional independence of matches is assumed; each observed opponent rank occurs in one home and one away fixture, so repeated opponents could introduce dependence. The data have no match dates or separate opponent identifiers, limiting checks of time dependence and opponent-specific effects. Final league rank is unavailable before the season ends. Model-selection uncertainty is not included in the reported intervals or p-values. These results support an exploratory description of this dataset, not a causal or externally validated claim.

Conclusion

The Week 6 selected model uses **home/away status and opponent rank** to explain goals scored. On all 46 matches it estimates **twice as many goals at home** and **5.6% more expected goals per one-place increase in numerical opponent rank**. Overall fit checks do not flag clear overdispersion, but the evidence for both associations is **sensitive to the 8-0 match**. Keep that observation and report the sensitivity alongside the main findings.

Reproducibility and sources

- `code/Week6_Analysis.R` was executed successfully in **R 4.5.0** using only standard R packages. It reads `data/ashford_matches.csv` and recreates four plots, five numeric tables and `outputs/week6_R_console_output.txt`.
- Figures and numbers in these notes come from that R execution. The full console output includes candidate fits, `drop1` and `anova` tests, all match diagnostics and `sessionInfo()`.
- Methods: *MTH6134-Week6-practical.pdf*, AIDS cases Questions 1-6 and Atomic bombs Questions 1-4. Week allocation: *MTH6134-data-brief.pdf*, page 2. The supplied *MTH6134-Weeks6and8-notes.pdf* motivates the subsequent contingency-table work.
- Continuity and layout: `week 2.R`, `Week 2 Notes.md`, `week4.R`, `Week 4 (league_players).md` and `Week3_Decision_Notes_Updated.md`. Earlier statements about diagnostics were checked rather than carried forward as assumptions.